CHAPTER 5
Magnetism and Matter

Magnetic Dipole Moment, and Force Between Magnetic Poles

1. The magnetic moment of an iron bar is M. It is now bent in such a way that it forms an arc section of a circle subtending an angle of 60° at the centre. The magnetic moment of this arc section is *(2024 Re)*

a. $\frac{3M}{\pi}$ b. $\frac{4M}{\pi}$ c. $\frac{M}{\pi}$ d. $\frac{2M}{\pi}$

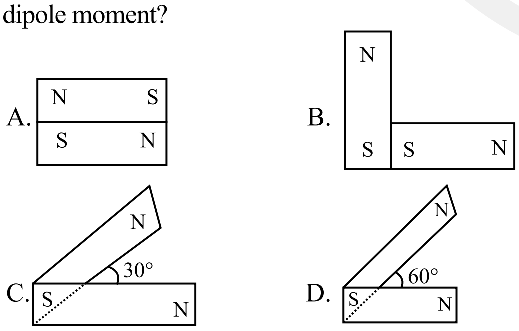
2. A sheet is place on a horizontal surface in front of a strong magnetic pole. A force is needed to: *(2024)*

A. hold the sheet there if it is magnetic.
B. hold the sheet there if it is non-magnetic.
C. move the sheet away from the pole with uniform velocity if it is conducting.
D. move the sheet away from the pole with uniform velocity if it is both, non-conducting and non-polar.

Choose the correct statement(s) from the options given below:

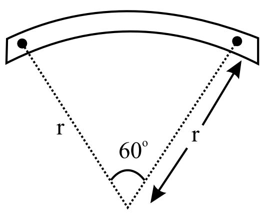
a. A, C and D only b. C only
c. B and D only d. A and C only

3. Following figures show the arrangement of bar magnets in different configurations. Each magnet has magnetic dipole moment m. Which configuration has highest net magnetic dipole moment? *(2014)*



a. A b. B c. C d. D

4. A bar magnet of length l and magnetic dipole moment M is bent in the form of an arc as shown in figure. The new magnetic dipole moment will be: *(2013)*



a. 2M b. M c. 3M/π d. π/2 M

Torque Acting on a Magnetic Dipole and Potential Energy of Dipole in a Magnetic Field

5. The magnetic potential energy, when a magnetic bar of magnetic moment $\vec{m}$ is placed perpendicular to the magnetic field $\vec{B}$ is *(2024 Re)*

a. $-\frac{mB}{2}$ b. Zero
c. $-mB$ d. mB

6. A 250 turn rectangular coil of length 2.1 cm and width 1.25 cm carries a current of 85 μA and subjected to a magnetic field of strength 0.85 T. Work done for rotating the coil by 180° against the torque is: *(2017-Delhi)*

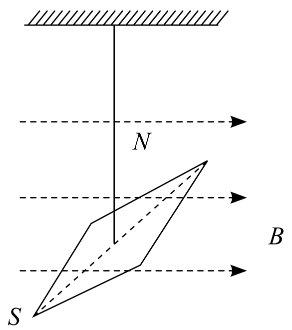
a. 4.55 μJ b. 2.3 μJ c. 1.15 μJ d. 9.1 μJ

7. A bar magnet is hung by a thin cotton thread in a uniform horizontal magnetic field and is in equilibrium state. The energy required to rotate it by 60° is W. Now the torque required to keep the magnet in this new position is: *(2016 - II)*

a. $\frac{\sqrt{3}W}{2}$ b. $\frac{2W}{\sqrt{3}}$
c. $\frac{W}{\sqrt{3}}$ d. $\sqrt{3}W$

Miscellaneous (Work Done by External Force in Magnetic Field and Vibration Magnetometer)

8. The magnetic moment and moment of inertia of a magnetic needle as shown are, respectively, 1.0×10^{-2} A m^2 and $\frac{10^{-6}}{\pi^2}$ kg m^2. If it completes 10 oscillations in 10 s, the magnitude of the magnetic field is *(2024 Re)*



a. 0.4 T b. 4 T c. 0.4 mT d. 4 mT

Magnetic Properties of Materials

9. The incorrect relation for a diamagnetic material (all the symbols carry their usual meaning and ε is a small positive number) is *(2024 Re)*

a. $\mu < \mu_0$
b. $0 \le \mu_r < 1$
c. $-1 \le \chi < 0$
d. $1 < \mu_r < 1 + \varepsilon$

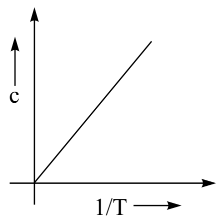
10. Match List-I with List-II. *(2024)*

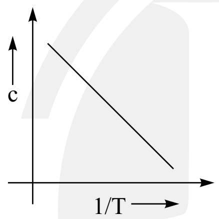
	List-I (Material)		List-II (Susceptibility (χ))
A.	Diamagnetic	I.	$\chi = 0$
B.	Ferromagnetic	II.	$0 > \chi \ge -1$
C.	Paramagnetic	III.	$\chi >> 1$
D.	Non-magnetic	IV.	$0 < \chi < \varepsilon$ (a small positive number)

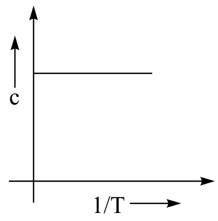
Choose the correct answer from the options given below:

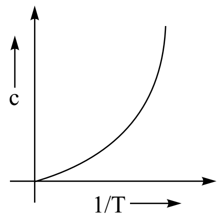
a. A-III, B-II, C-I, D-IV
b. A-IV, B-III, C-II, D-I
c. A-II, B-III C-IV, D-I
d. A-II, B-I, C-III, D-IV

11. The variation of susceptibility (χ) with absolute temperature (T) for a paramagnetic material is represented as: *(2023-Manipur)*

a.


b.


c.


d.


12. An iron rod of susceptibility 599 is subjected to a magnetising field of 1200 A m^{-1}. The permeability of the material of the rod is : *(2020)*

($\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1})

a. 8.0×10^{-5} T m A^{-1}
b. $2.4\pi \times 10^{-5}$ T m A^{-1}
c. $2.4\pi \times 10^{-7}$T m A^{-1}
d. $2.4\pi \times 10^{-4}$ T m A^{-1}

13. A thin diamagnetic rod is placed vertically between the poles of an electromagnet. When the current in the electromagnet is switched on, then the diamagnetic rod is pushed up, out of the horizontal magnetic field. Hence the rod gains gravitational potential energy. The work required to do this comes from: *(2018)*

a. The lattice structure of the material of the rod
b. The magnetic field
c. The current source
d. The induced electric field due to the changing magnetic field

14. The magnetic susceptibility is negative for: *(2016 - I)*

a. Diamagnetic material only
b. Paramagnetic material only
c. Ferromagnetic material only
d. Paramagnetic and ferromagnetic materials

Answer Key

1. (a) **2.** (d) **3.** (c) **4.** (c) **5.** (b) **6.** (d) **7.** (d) **8.** (c) **9.** (d) **10.** (c)
11. (a) **12.** (d) **13.** (c) **14.** (a)